

Secure-FEPRH: Ultimate Learning Resource Guide

This document contains the absolute best, hand-picked YouTube tutorials and crash courses for every single technology used in the Secure-FEPRH project. Share this with your team so everyone can master their specific module!



1. The Frontend Stack (React, Vite, Tailwind CSS)

For the student handling UI, UX, and the frontend foundation.

- **React JS Full Course 2023 | Build an App and Master React in 1 Hour** (*JavaScript Mastery*)
 - **Why:** The best modern crash course on React components, hooks (`useState` , `useEffect`), and props using Vite.
 - **Tailwind CSS Full Course for Beginners** (*freeCodeCamp.org*)
 - **Why:** A complete 3-hour deep dive into building beautiful, responsive UIs with Tailwind CSS, exactly like the ones used in our National Command Center dashboard.
 - **Framer Motion Crash Course** (*Net Ninja*)
 - **Why:** Learn how to add the smooth micro-animations and page transitions we use in the layout.
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2. The Core Backend Stack (FastAPI, Python, PostgreSQL)

For the student (You) handling the core architecture and database.

- **FastAPI Tutorial - Build a REST API with Python** ([freeCodeCamp.org](#) / [Sanjeev Thiyagarajan](#))
 - **Why:** An incredible 19-hour masterclass that covers *everything*—FastAPI, PostgreSQL, SQLAlchemy, JWT Authentication, and Alembic migrations. If you watch one thing, watch this.
 - **FastAPI SQLAlchemy 2, Alembic and PostgreSQL Setup Tutorial** ([Tenacity Dev](#))
 - **Why:** A shorter, 2025-updated guide focusing specifically on hooking up SQLAlchemy 2.0 with a Postgres database cleanly.
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3. Security & Advanced Backend (JWT, RBAC, Crypto)

Also for the core architect.

- **JWT Authentication in FastAPI** ([Indently](#))
 - **Why:** Explains how JSON Web Tokens work and how to implement Role-Based Access Control (RBAC) securely in FastAPI.
 - **Python Cryptography Tutorial - Fernet & AES** ([NeuralNine](#))
 - **Why:** Essential for understanding how the Secure Repository encrypts and decrypts files before saving them to the disk.
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4. Artificial Intelligence & Biometrics Engine

For the student handling the Plagiarism and Attendance AI.

- **Face Recognition with Python and OpenCV** ([Murtaza's Workshop](#))
 - **Why:** A brilliant, easy-to-follow guide on using the `face_recognition` library to encode faces into 128D arrays and compare them for attendance tracking.

- **Natural Language Processing (NLP) in Python - TF-IDF and Cosine Similarity**

(freeCodeCamp.org)

- **Why:** Explains the exact math and code needed to compare project abstracts for text plagiarism.

- **Abstract Syntax Trees (AST) in Python** (AnthonyWritesCode)

- **Why:** Understand how to parse Python code into a tree structure to detect structural code plagiarism (bypassing variable renaming).



5. Asynchronous Background Tasks (Celery & Redis)

For handling heavy loads without freezing the server.

- **Background Jobs with Celery, Redis & Docker** (Very Academy)

- **Why:** Shows how to offload heavy tasks (like nationwide plagiarism scans) to a Celery worker using Redis as the message broker.



6. Real-Time Collaboration (WebSockets)

For the student building the Kanban board.

- **FastAPI WebSockets Tutorial** (CodingWithRoby)

- **Why:** Learn how to open a two-way connection between React and FastAPI to make tasks move across the Kanban board instantly for all users.

- **Drag and Drop with React Beautiful DnD / @hello-pangea/dnd** (freeCodeCamp.org)

- **Why:** The exact frontend logic needed to create the interactive drag-and-drop columns for the Kanban board.